

ÉCOLE POLYTECHNIQUE FÉDÉRALE DE LAUSANNE



Low-Cost Raman Microscope



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1 Introduction

Plastic pollution, especially in the form of microplastics, represents a major and growing threat to marine ecosystems and human health. Microplastics, which are plastic particles smaller than 5 mm, are now present in all oceanic environments, from deep-sea sediments to surface waters. Because of their small size, a variety of marine creatures, including fish, sea turtles, and plankton, can consume them. This could lead to the spread of plastic particles and related poisons up the food chain and ultimately to people. According to some research, microplastics may interfere with marine animals' ability to reproduce, metabolize, and respond to immunological stimuli. Their long-term impacts on human health are still being studied.

Despite the urgency of the problem, detecting and analyzing microplastics *in situ* remains a major technical challenge. Most high-precision instruments for identifying plastic types, such as Fourier Transform Infrared (FTIR) or Raman spectrometers, are expensive, cumbersome and generally confined to laboratory environments. Field research, particularly aboard small or low-tech vessels such as sailboats, requires equipment that is not only accurate, but also compact, energy-efficient, affordable and resistant to harsh environmental conditions such as humidity, vibration and movement.

Raman spectroscopy is particularly well suited for this application. Unlike FTIR, Raman techniques are less affected by water interference and require minimal sample preparation. The Raman effect provides a unique spectral fingerprint for each material, allowing for accurate identification of plastic types even in complex or contaminated samples. Moreover, Raman spectroscopy can be integrated into compact optical systems, making it a strong candidate for portable or low-cost implementations in marine environments.

2 Project description

This project aims to develop a low-cost Raman microscope capable of identifying microplastics in samples collected *in situ*. The microscope is intended to operate in field environments where traditional laboratory equipment is impractical due to cost, size or complexity. This project is being carried out in partnership with the Sailowtech association and the LNET laboratory.

The device should combine optical microscopy and Raman spectroscopy to enable both visual observation and chemical characterization of microparticles. Raman spectroscopy is particularly well suited to this application, as it enables non-destructive analysis with minimal interference from water, making it ideal for marine samples.

2.1 Project Context

Sailowtech is an association of EPFL students whose aim is to develop a more sustainable approach to scientific research. The aim is to rethink our use of resources and energy in scientific research. On campus, students develop low-tech devices that can be used on the sailboat, and in parallel, often an expedition is led by a few people, including students, where they can test what has previously been built on campus. During these expeditions, they also mainly carry out research into aquatic environments. So, with this project, Sailowtech wants to be able to identify the microplastics present in the samples they collect during their expedition. This will provide data on the microplastics present in different parts of the ocean.

The LNET laboratory is located at EPFL and has a branch specializing in optics. Thanks to their knowledge and equipment, we were able to carry out this project in the best possible way.

The project did not start from scratch, but is in fact a continuation of the project led by Louis Litzler-Italia, Alan Reuvekamp and supervised by Dr. Alan Bowman. They were also both working on the development of a Raman microscope with the sailowtech association and the LNET laboratory. We picked up the project where they left off, identified several limitations in the initial prototype, and undertook a redesign to improve performance and usability.

This project was carried out during the semester by Francesco and myself. At the end of the semester, two additional students joined the team, and they are now continuing the development of the microscope.

2.2 Low-Tech

We should start by defining the word “low-tech”, in order to characterize the project a little more precisely. There are in fact many different definitions, but here's a fairly comprehensive one.

A low-tech technology is one that is based on these three main principles[1]:

- *Useful*: A low-tech product meets essential individual or collective needs. It contributes to making healthy lifestyles, production and consumption possible and relevant for all.
- *Accessible*: Low-tech products must be accessible to as many people as possible. It must therefore be possible to manufacture and/or repair it locally, its operating principles must be easy to understand, and its cost must be affordable for a large part of the population.
- *Sustainable*: Eco-designed, resilient, robust, repairable, recyclable, agile, functional: low-tech invites us to think about and optimize the ecological, social and societal impacts of using technologies at every stage of their lifecycle (design, production, use, end-of-life).

3 Specification

The aim of this project is to develop a portable, low-cost, low-tech Raman microscope that can be taken on a sailing expedition to identify microplastics on the spot. To achieve this, we can divide the specifications into two parts: on the one hand, the environmental constraints linked to the sailboat environment, and on the other, the user's requirements, in order to simplify use of the device as much as possible.

Environmental Constraints:

- *Water resistance and moisture protection*: The device must be sealed to prevent water penetration and protect internal components from moisture.
- *Power supply*: It must run efficiently on a 12 V battery, consuming a minimum of power to guarantee prolonged use.
- *Resistance to vibration and movement*: The unit must resist vibration and boat movement, including pitch and roll, without compromising performance.

User Requirements:

- *Cost-effectiveness*: The device must be affordable, and prioritize essential functions over high-end options.
- *Portability*: It needs to be compact and lightweight for easy transport and handling.
- *User-friendly interface*: The interface should be intuitive and require minimal training or prior knowledge.
- *Accuracy of microplastics analysis*: The device must accurately analyze microplastics between 2 and 50 micrometers in size.
- *Dual function*: capture high-quality microscopic images and perform spectrometric analysis.
- *Spectral resolution*: The device must have a spectral resolution of less than 5 cm^{-1} for accurate measurements.

4 Working Principle of Raman Spectroscopy

4.1 Light Behavior

Before discussing the operating principle of Raman spectroscopy, it's useful to understand how light behaves when it encounters matter. Several interactions can occur, as illustrated in Figure 1.

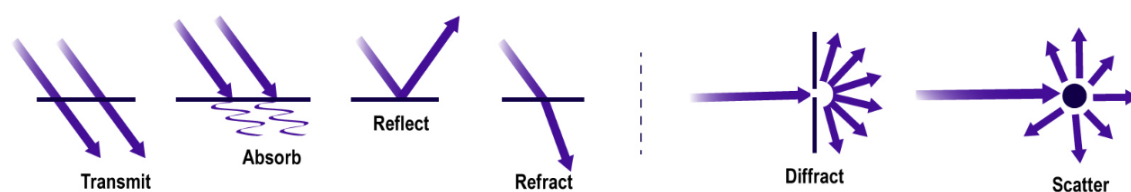


Figure 1: Examples of the ways light moves and behaves as it interacts with a medium (left), such as the atmosphere, the ocean or a small object like an aperture or particle (right). [2]

In the context of Raman spectroscopy, we focus only on interactions occurring at the molecular level, rather than on large-scale phenomena in bulk media. Specifically, we are interested in three key behaviors: **transmission**, **absorption** and **scattering**. These interactions form the basis of how light interacts with molecules, and will help us to better understand Raman spectroscopy.

- Transmission does not affect the molecule; light simply passes through it.
- Absorption occurs when light strikes the molecule and causes it to vibrate. This vibration generates heat and therefore energy, and if enough energy is absorbed, some may be re-emitted in the infrared (Fluorescence).
- Scattering occurs when light strikes an object and bounces back in many directions.

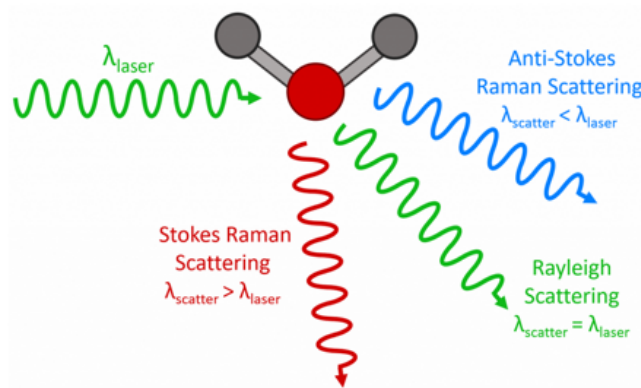


Figure 2: Molecule hit by a monochromatic light [3]

When the light is scattered, two phenomena can actually happen, as shown in Figure 2:

- Elastic scattering (also called Rayleigh and Mie scattering): when the scattered light has the same energy as the incident light.
- Inelastic scattering (also called Raman scattering): a photon exchanges energy with the vibrational states of the molecule.
 - If the emitted photon has lower energy than the incident one, we observe a Stokes shift.
 - If the emitted photon has higher energy than the incident one, we observe an anti-Stokes shift.

Note that Figure 2 is based on wavelength rather than energy. But it's actually the same thing, since energy is inverted in proportion to wavelength, as it can be seen in the following formula: $E = h\nu = \frac{hc}{\lambda}$, with h Planck's constant and c the speed of light. For example, for Stokes Raman Scattering we have as shown in the figure:

$$\lambda_{scatter} > \lambda_{laser} \Leftrightarrow \frac{hc}{E_{scatter}} > \frac{hc}{E_{laser}} \Leftrightarrow E_{scatter} < E_{laser}$$

Thus, during Raman Stokes scattering, the scattered photon has a lower energy than the incident photon (laser).

4.2 Raman Spectroscopy

The aim of Raman spectroscopy is to identify any microplastics present in the given sample. To achieve this, several steps are necessary, as illustrated in Figure 3

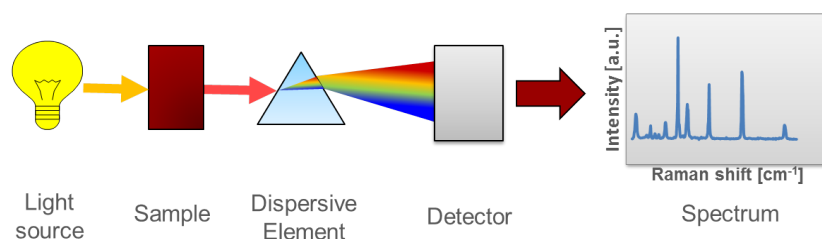


Figure 3: Diagram showing the operating principle of a Raman spectrometer[4]

First, the sample is illuminated by monochromatic light (usually a laser). When the laser comes into contact with the sample, most photons are either transmitted through the material or elastically scattered. This elastic scattering contains no molecular information. However, in very rare cases, around 1 in $10^6 - 10^8$ photons, the photon is inelastically scattered (a process known as Raman scattering). Inelastic scattering causes a slight energy shift. This energy shift is highly specific to molecular bonds and structure, making the Raman spectrum a unique chemical fingerprint of the sample. In some cases, the photon can be absorbed and often re-emitted as infrared (see Figure 4), a phenomenon known as fluorescence. We want to avoid this phenomenon, as it is stronger than the Raman signal and interferes with it[5].

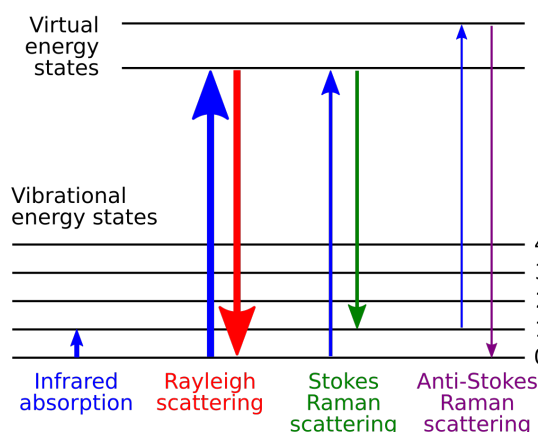


Figure 4: Diagram of energy levels involved in spectroscopy, the thickness of the lines indicates qualitatively the signal intensity of each transition. [6]

The scattered light then passes through a long-pass optical filter, not shown in Figure 3 but located between the sample and the dispersive element, which blocks the intense Rayleigh component, allowing only the weaker Raman light to reach the next stage.

This filtered light is directed onto a dispersive element, such as a diffraction grating, which separates the Raman-scattered light according to wavelength. A detector then records the intensity of each wavelength to create a spectrum (see Figure 5). This spectrum displays peaks that correspond to the molecular vibrations of the sample, providing a unique molecular identification. By analyzing the Raman shift as a function of intensity in this spectrum, we can match it to a known database of Raman spectra, enabling us to identify the microplastics present in the sample.

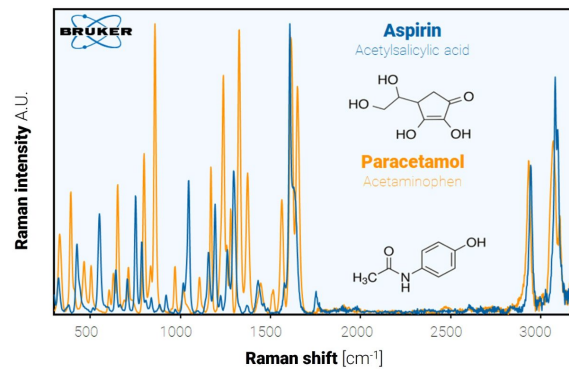


Figure 5: Example of Raman Spectrum[7]

Raman spectroscopy has some advantages which make it ideal for our application :

- **Non-destructive:** Raman spectroscopy does not damage the sample, making it ideal for reuse or preservation of the sample.
- **Sample diversity:** samples can be in the form of solids, liquids, or gases.
- **Works through transparent containers:** Raman measurements can be carried out through glass or plastic vials to avoid contamination or exposure.
- **Water compatibility:** Raman spectroscopy is ideal for analyzing aqueous samples because, unlike infrared (IR) spectroscopy, the Raman signal from water is weak.

In this project, we aim to build a Raman microscope, an object that combines Raman spectroscopy with an optical microscope to analyze very small sample areas with high spatial resolution.

5 Summary of the Initial Designs

As we said earlier, the project wasn't started from scratch, but we took over from a previous version. In this section, we'll look at the initial setup configuration. In Figure 6, we have an overview of the initial design of the Raman microscope.

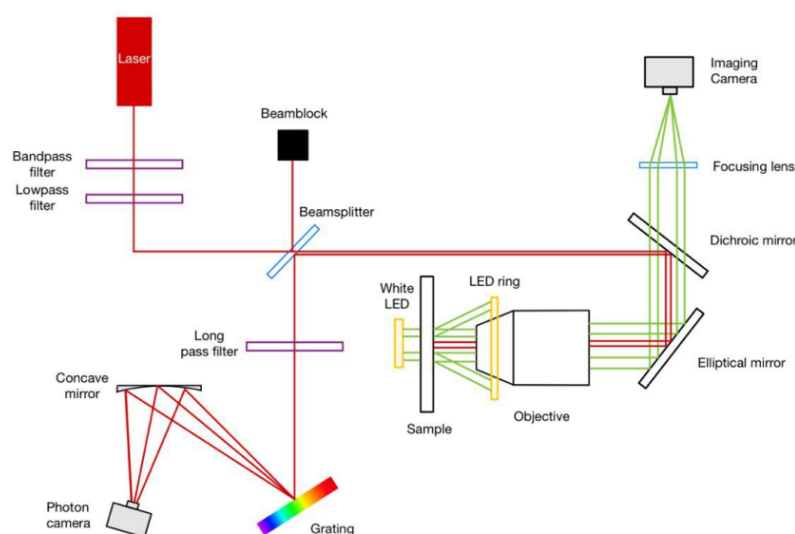


Figure 6: Initial Design of the Raman Microscope[8]

The green rays represent the white light used in an optical microscope, while the red rays represent the spectroscopy part. Let's start with the optical microscope. The sample can be illuminated by an

LED in either a reflective or a transmissive LED. The light then passes through an objective lens (The objective is the 40X ZEISS A-Plan Objective) to collimate the beam, as in a normal microscope. After passing through a focusing lens, it is collected by an imaging camera. This camera enables us to obtain an image of the sample. The principle is exactly the same as that of a normal microscope. The only difference is that the light passes through an elliptical mirror and a dichroic mirror. These mirrors are only there for geometric reasons and do not affect how the microscope operates. They are there so that we can have both an optical microscope and a spectrometer. The dichroic mirror will have only a slight effect on the microscope, since, in transmission, it only allows transmission of wavelengths below 600 nm. This means that the image of the sample collected will appear slightly blue/green as the red is all reflected. However, this doesn't affect operation at all, since we're only interested in the shape of the microplastic sample and not its color.

For the spectroscopy part, the sample will be illuminated by monochromatic light, in this case a laser at a wavelength of 639 nm. This light is filtered to ensure that only the laser wavelength reaches the sample. Once it reaches the sample, Rayleigh and Raman scatterings occur. On the way back, the reflected and scattered light passes through the objective, which is infinity-corrected, so that from this point onwards, the beam should be collimated. The dichroic then reflects only the wavelength above 600 nm, the beam passes through the beam-splitter, and on through the long-pass filter. This long-pass filter blocks laser light below 650 nm, and thus Rayleigh scattering. Only Raman scattered light reaches the grating. The grating is our dispersive element, an optical element that reflects each wavelength at a different angle, as shown by Figure 10. Once it has passed the grating, it's no longer a beam but each wavelength at a different angle, which is why we have a concave mirror, to collect all the wavelengths and then focus them together so that they can all enter the photon camera chip and allow us to collect the intensity of each wavelength.

The imaging camera was controlled by a Raspberry Pi, but the photon camera required a PC for operation.

In the previous setup, only the optical microscope could be tested, not the spectroscopy setup, because the chosen grating was unsuitable. The first step was therefore to select a new grating that was suitable for our device.

6 Overview and Comparison of Existing Low-Cost Raman Setups

Low-cost Raman microscopes have already been developed, and some are already on the market. We'll take a look at the different systems.

6.1 OpenRaman

OpenRaman is a low-cost, entirely open-source Raman microscope, costing around 2,500 CHF for the basic version. We took inspiration from it to find a solution for the calibration of our photon camera, since we have the same camera as them.

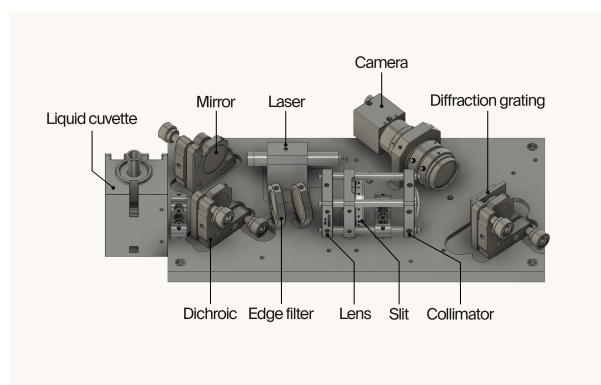


Figure 7: Diagram of the OpenRaman microscope[9]

The biggest difference with our device is that they don't use a concave mirror, but a lens on the camera, to be sure of collecting all wavelengths anyway. They also use a different system before the

In general, most systems use a slit before the grating. A slit is generally used to control spectral resolution; the smaller the slit, the better the spectral resolution, but the smaller the slit, the less light will pass through and the less Raman signal will be detected.

OpenRaman therefore uses a very intelligent system, a telescope. This allows them to have a larger slit, and therefore collect more light, while maintaining good spectral resolution since, through the telescope, the beam size is increased and hits the grating with a large beam. There is no interference, as the beam is fully colimated by the telescope[10]. The result is a resolution of 12 cm^{-1} [9].

6.2 Low-cost Spectrometer

This is a low-cost spectrometer based on the same principle as our spectrometer, except that it uses a linear detector instead of a photon camera (see Figure 8). As the linear detector is 1D, alignment is easier. The original design of our Raman microscope included a linear detector, but Louis modified it due to lack of documentation, flexibility and reliability, as he describes in his report[8].



For the rest of the installation, they use the same elements as we do, a grating and a concave mirror. All the optical elements are attached directly to a 3D-printed breadboard, which is much easier for alignment, but the disadvantage is that some elements might move as the plastic can deform over time. Using this spectrometer with a 639 nm laser, the resolution would be 1.27 cm^{-1} . This is actually quite good, since Raman microscopes on the market have a resolution of around 1 cm^{-1} [12]. But even the designer of this spectroscopy isn't sure that this data is entirely accurate.[11].

The RamanPi is a low-cost, fully automated Raman microscope running on a Raspberry Pi. It also uses, as the spectrometer seen above, a light detector instead of a photon camera. It is entirely 3D printed and designed so that the optical elements are attached to the 3D printed parts and to each other. (see Figure 9). During measurement, the filters are changed automatically to take measurements with both, then the collected spectra are superimposed to obtain the best possible spectrum whatever the wavelength range associated with the microplastic. The spectral resolution of RamanPi is around 10-20 cm^{-1} [13].

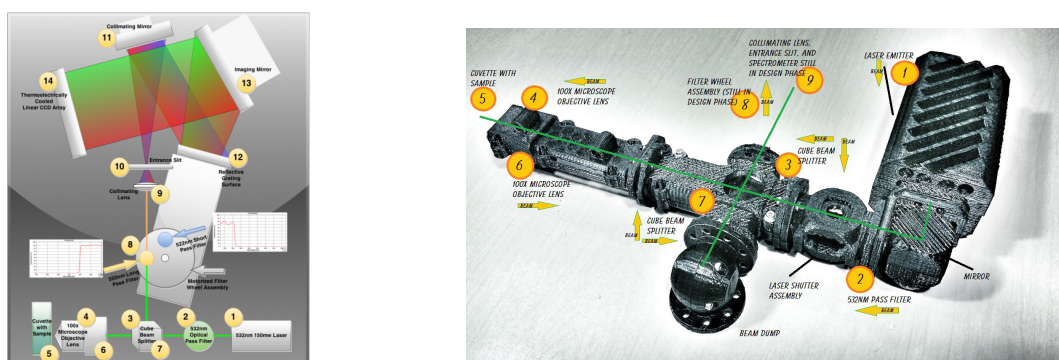


Figure 9: Diagram and Photograph Illustrating the RamanPi Device [13]

7 Diagnosis and Identified Issues

7.1 Grating

The grating is a very important part of the Raman microscope. When light hits the grating, it diffracts and constructively interferes at specific angles depending on its wavelength. This phenomenon is described by the grating equation :

$$n\lambda = d \cdot (\sin(\alpha) - \sin(\beta)) \quad (1)$$

where:

- n is the diffraction order.
- λ is the wavelength of light.
- d grating spacing (inverse of grooves per mm).
- α is the angle of incidence.
- β is the angle of diffraction.

For Raman spectroscopy, we're only interested in the first order ($n=1$) because it's the first order of diffraction and allows us to see a clear spectrum, with sufficient dispersion and intensity. Order 0 is in fact just a reflection of the incident beam (see Figure 10), and the higher orders are more dispersed, which would facilitate wavelength collection since they are more widely spaced, but the intensity is very low, so we can't use it for Raman spectroscopy.

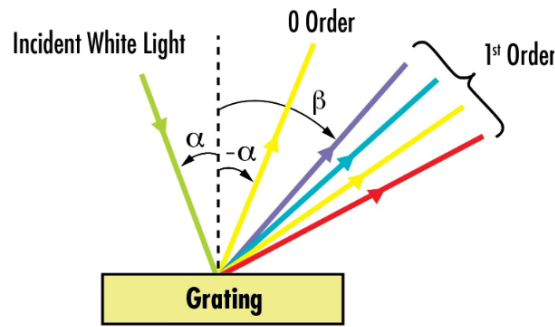


Figure 10: Diffractions Orders Grating [14]

In Raman spectroscopy, the spacing and angle between the grating, the concave mirror and the photon camera are crucial, as we need to capture all the wavelengths of interest for the Raman signal with the photon camera. However, the wavelengths must be far enough apart to be distinguishable from one another. As can be seen in Equation 1, the β angle depends on the groove density of the grating (see Figure 11). In the previous design, the groove density was 1200 grooves/mm. This presented a problem because, with the configuration we had, if we wanted to capture all the wavelengths we were interested in with the concave mirror, the grating and the mirror would have had to be spaced 3.3 mm apart. This was physically impossible.

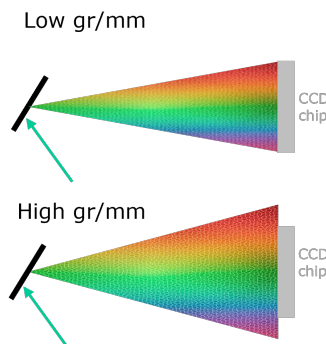


Figure 11: Grating with different groove densities [15]

Knowing that we didn't want to modify the set-up too much, i.e. we always wanted to stay within the reasonable range of the Raman shift (200 cm^{-1} to 3400 cm^{-1}), keeping the laser at the same wavelength (639 nm) and always capturing the first order of diffraction, the simplest parameter was to modify the density of the grating grooves.

Input	Output
Raman shift Range	Wavelength Range
Diffraction order	Invalid value of α
Grating groove/mm	Min and max β
Blaze angle	Grating to mirror distance
Incident angle α	Mirror to detector distance
Photon camera chip size	
Concave mirror focal length	

Table 1: Calculation parameter for grating selection

Table 1 describe the parameters needed to correctly select the grating. The grating density is in fact an input and not an output, as we have calculated it the other way round. In fact, there aren't many choices of grating density (300, 600, 830, 1200,...), so we made the calculation for these different groove densities and then observed which distance and angle was the most appropriate for the configuration. Francesco transferred all these calculations to a Python code to simplify the calculations. The result is that the most appropriate groove density is 600 grooves/mm. The distance between the grating and the mirror should then be around 60 mm, which is much better than 3.3 mm. As the angles are entered but within a range, we have for each different angle (between 11 deg and 15 deg for 600 grooves/mm) the rest of the corresponding distances.

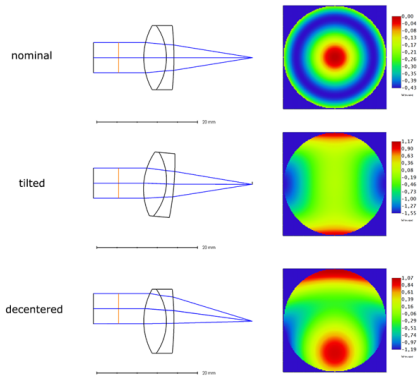
7.2 Optical Alignement

Once we had purchase the new grating, we were finally able to test our installation. We started on the lab's optical table.



Figure 12: Raman Microscope on the bench lab

Being new to optics, we had a lot of trouble with the alignment at first. We spent hours doing the first alignment and it still wasn't good enough, but it got easier by the second alignment. At some point we realized that most of our problems trying to do the Raman function came partly from the fact that the alignment wasn't precise enough. One thing we've discovered over the semester is that with a powermeter (see Figure 13b), we can easily spot those misalignments that escape us with the naked eye. In fact, a small lens that's tilted or off-center can result in a significant loss of power, as shown in Figure 13a. Ultimately, we were losing the entire Raman signal, since it's already very weak.

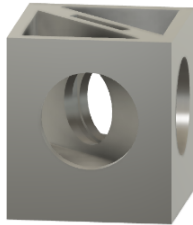


(a) Power loss due to different lens misalignment



(b) Powermeter from ThorLabs

To simplify the alignment process, the 3D-printed mounts designed by Louis (see Figure 14a) have been replaced with a commercial alternative. Once the system is up and running, these mounts can be integrated to make the microscope more accessible and affordable.



(a) 3D printed mount for the dichroic mirror

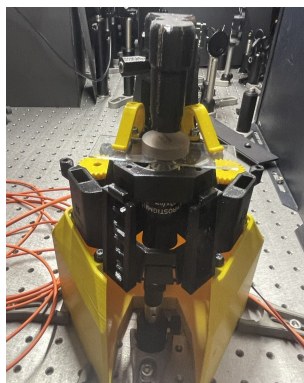


(b) Commercial mount from ThorLabs

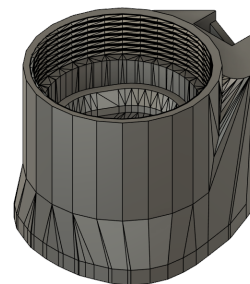
Figure 14: Holders of optical element

7.3 Microscope Focus

The microscope we're using is the open-source microscope from OpenFlexure [16] with a few modifications made by Louis to adapt it to a Raman microscope, and thus be able to have a spectroscopy part (see Figure 15a). The modifications are well described in Louis's report [8]. One problem we encountered when testing the Raman microscope was the adjustment of the optical microscope. In fact, we could no longer focus the microscope. The objective couldn't move up enough on the z-axis to get the right focal length. This was a major problem, because without focusing on the sample, the rays on the return path are not infinitely collimated and therefore not parallel. We absolutely need parallel rays to reach the grating, otherwise interference will appear in the reflected wavelength since the angle of incidence was not the same for all rays.



(a) OpenFlexure Microscope Modified



(b) Objective holder, part of the microscope, Fusion design

The easy solution found to this problem was to modify the height of the lens holder, illustrated in Figure 15b, so that it was already high enough to focus on the sample. Several different holders were 3D printed before the right height was found.

During the operation, we realized that the z-axis O-ring had broken. The openFlexure microscope works on a very clever principle. The central part is a 3D printed part with no support. To move on the axis, the principle is quite simple: simply turn a wheel as for a conventional microscope, one side will push on the plastic and, by simple deformation of the plastic, the axis will move. If the wheel is turned in the opposite direction, an elastic O-ring will pull on the plastic and, by deformation, also move the axis. This allows each axis to move in both directions, thus increasing the range.

The problem is that these O-rings aren't easy to find - you have to order them from a very specific online store (so it's not exactly low-tech) - and they don't seem to stand the test of time. In fact, the O-rings already on the microscope were installed less than a year ago, and the z-axis had already broken. The x and y axes were almost as broken due to age, and could be seen cracking. For a finished portable Raman microscope, this is certainly something that needs to be changed, or at least thought about, because it's not durable and would need to be changed often. In the microscope's current state, changing them means dismantling the whole microscope to replace them. On a finished product, this should be avoided as far as possible, to minimize the risk of element movement and misalignment.

Another idea would be to remove them altogether. The plastic would then only be able to deform in one direction (by pushing), but by adjusting certain parts, as was done with the z-axis attachment, this could work. Again, the question is whether this will have any impact on microscope operation, ease of use for the user, etc.

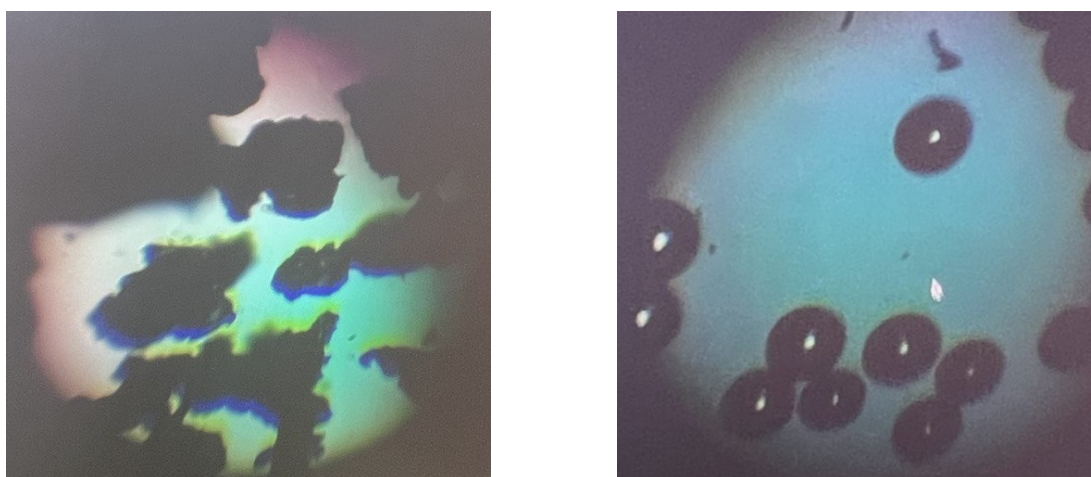


Figure 16: Microplastic sample seen through the microscope

7.4 Photon Camera Calibration

The photon camera is the Flir BFLYPGE Blackfly® PoE GigE monochrome camera. It is in fact a camera adapted to Raman spectroscopy as used in the OpenRaman microscope[9].

The camera captures the scattered light after it has passed through the grating. Each column of pixels on the camera chip corresponds to a wavelength. The camera then records the amount of light collected at each wavelength, which corresponds to the intensity of each wavelength. With this data, the OpenRaman software linked to the camera is able to create a spectrum with the pixel column as a function of intensity. To associate each pixel row with a wavelength or Raman shift, the camera actually needs to be calibrated, using a neon or argon lamp. The software already knows the spectrum of these lamps, so by placing the lamp in the usual laser path, we should be able to collect the spectrum of the lamp for this given position of the installation. The software will then match the peaks obtained on the camera chip to the neon spectrum to determine which row of pixels corresponds to which wavelength. If any part of the system is moved, the calibration must be repeated, as the beam position will no longer be exactly the same, and the pixels columns will no longer correspond to the correct wavelength.

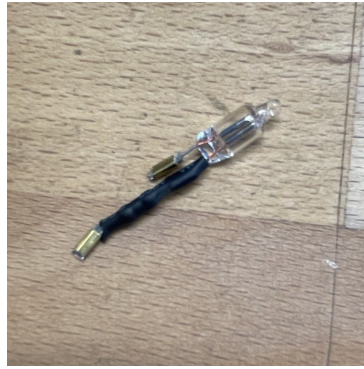


Figure 17: Neon lamp used for calibration

Unfortunately, LNET doesn't have an argon or neon lamp for optical calibration. As these are very expensive devices, we tried to find another solution. In OpenRaman, they actually use a small neon lamp, so we also tried to find one. On the Figure 17 is the neon lamp found at the waste disposal site, which we use for calibration after wiring it up to plug into the mains.

Once this neon lamp was found, the process of calibrating the camera could begin. We still haven't managed to do it. In the next section, I'll describe all the problems encountered and solutions for trying to calibrate the camera.

- **Misalignment:** significant power loss, resulting in no Raman signal on the camera chip.
- **Placing the neon lamp on the path:** The neon lamp was first placed where the sample was to be, but in doing so, the neon light passed through the dichroic mirror. The dichroic mirror reflects all wavelengths above 600 nm, whereas neon has a strong peak below 600 nm. This placement prevented us from obtaining the full spectrum of neon. The neon is now placed just after the dichroic mirror so that it can still cover most of the distance covered by the laser.
- **Uncollimated neon beam:** By placing the neon lamp after the dichroic, the neon light was not collimated, as it did not pass through the objective.
- **Not centered on the photon camera chip:** The basic setting of the photonic camera collects only a small part of the center of the chip (on the z axis). This explains why so many spectra were collected with an intensity comparable to noise, since our beam was not exactly centered on the camera's z-axis, but always a little lower or a little higher.
- **Incorrect grating angle:** The grating angle was too large to start with, leading to non-separated wavelengths on the chip camera.

Initially, the spectroscopy was aligned with the laser only, which led to an alignment error. As the laser is monochromatic, the grating reflected only one wavelength. That's why it took us so long to realize that the wavelengths were too close together, so the camera couldn't distinguish between them. To solve this problem, the neon lamp was used to align the spectroscopy part. In fact, it enabled us to see all the colors visible after the grating. Spinview software was also used. This software allows access to the image part of the photonic camera.

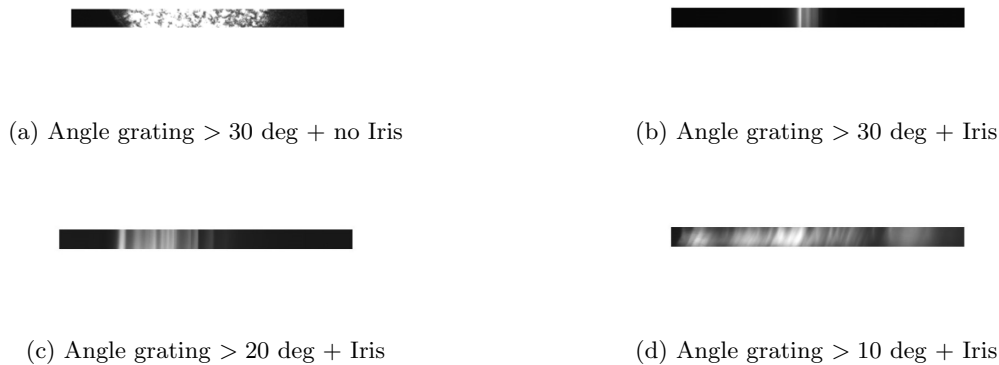


Figure 18: Image of Neon lamp on the photon camera collected via SpinView

Figure 18a shows that without an iris to reduce beam size/power, rays cannot be distinguished on the photon camera. An iris was therefore used, as the beam is too wide. it's what we had on hand, but it might be even better to use a slit or pinhole. Figure 18b shows that the addition of an iris produces sharp rays. With Figure 18b, Figure 18c, Figure 18d, we can see the importance of the grating angle. If the rays are too close together, the camera is unable to distinguish them. The aim is to use the entire camera chip to achieve better resolution.

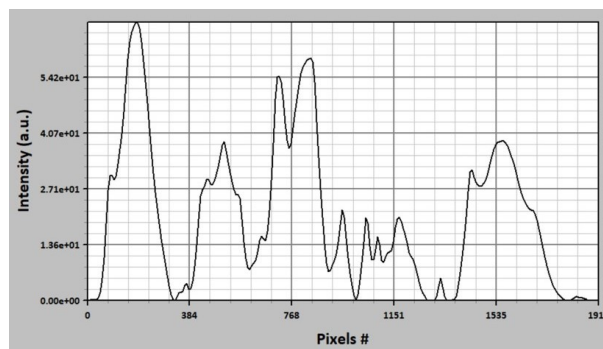


Figure 19: Pixel row vs. intensity for the image in the Figure 18d with OpenRaman

Figure 19 gives us the spectrum of the image in Figure 18d. This is quite promising, since the intensity is high enough not to be mistaken for noise. The peaks are relatively sharp, although this could be improved by first obtaining a sharp image in SpinView. The one we have is a little blurred, which explains the lack of sharpness of the peaks.

This calibration is still in progress, as at the time we had not yet succeeded in obtaining a good neon spectrum and therefore a good calibration. For the time being, new options are being explored, such as using a slit or even a telescope. It's definitely a subject that needs to be explored further, to be able to make a good calibration but also a good prototype to make this calibration easily as many times as necessary.

8 Enclosure Concept for Field Use

Right from the start of the project, the portable aspect of the Raman microscope was taken into account. As soon as the lab bench set up was working, the idea was to transfer it. Unfortunately, this never happened, but plans were made for when it does.

8.1 Breadboard Design

The first step was to find a suitable breadboard to transfer the device to.

For reasons of cost and accessibility, 3D printing was initially considered. At the time, the total dimensions of the optical part were not exactly known, but the approximate dimensions estimated were

500 x 400 mm. This posed design problems from the outset. In fact, the ordinary EPFL 3D printer, which most facilities have available, has a maximum printing area of 250 x 210 mm. The Prusa XL printer was also considered, but even that is too small, with dimensions of 360 x 360 mm. The solution was to create multiple parts that could be easily assembled, as shown in Figure 20. The idea was to be able to add an element if we needed more space on the breadboard.

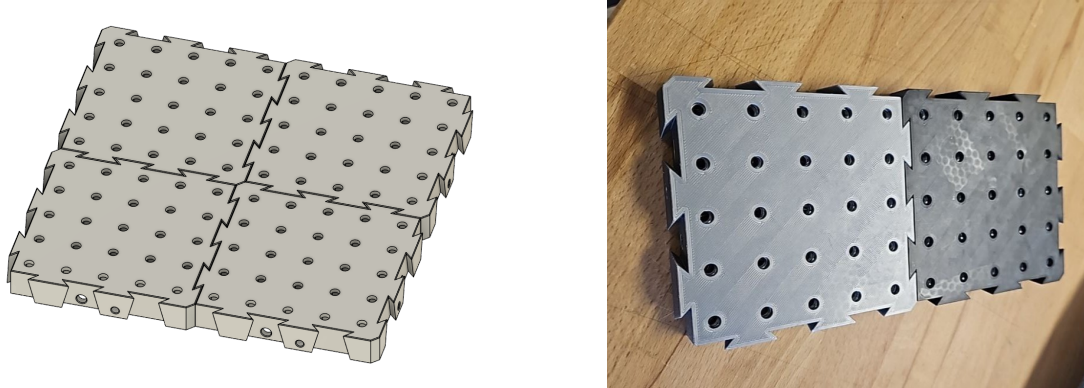


Figure 20: 3D printed Breadboard

Inside the breadboard are bolts, so that the optical element can be screwed in properly. The bolts were chosen against inserts and 3D-printed thread, for the precision and durability of the product. Indeed, 3D-printed threading is not always precise and can lead to the screw loosening, etc. The insert, when assembled, can be slightly tilted and this can lead to a lot of inaccuracy for optical alignment. To insert the bolts into the breadboard, 3D printing was stopped in the middle, and only resumed once all the bolts had been manually inserted into the breadboard.

After testing and realizing that the optical alignment had to be very precise and not move at all, the 3D printing option was abandoned. In fact, between two breadboards they moved a little, which was already too much.

The second option considered, was to manufacture our own aluminium breadboard at EPFL's SPOT, on the metal CNC machine they have, as shown in Figure 21



Figure 21: Milling machine SPOT

Unfortunately, this option was not viable for a number of reasons:

- Machine's maximum milling surface was 300 x 250 mm
- Raw material was as expensive as buying a commercial breadboard
- Not as precise as commercial breadboard

In the end, the best option for the time being was to buy a commercial breadboard from ThorLabs. For this, the total dimensions of the optical elements were needed.

Initially, the idea was to run the Raman on the laboratory bench, and then, directly on the bench, to optimize it as best. As the Raman never worked, a 3D model was made on Fusion to get a better idea

of the dimensions. To achieve the best possible approximation, all holders and mounts are exactly the same as those used in the actual installation.

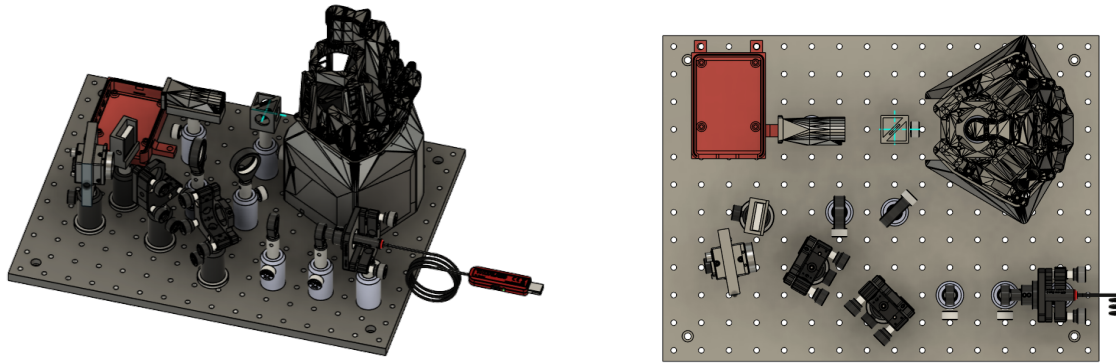


Figure 22: Preliminary Fusion model of the Raman

On this Figure 22, each optical component is as close as possible, leaving a little clamp space (not in the figure) for certain elements that need to be oriented at an angle. Each element has its definitive place, with the exception of the grating, the concave mirror and the photon camera, which still need to be adjusted during calibration. The total dimension of the plate in this Figure 22 is 400 x 300 mm. After consulting the ThorLabs website, the most suitable plate they suggest is 450 x 300 mm. The extra 5 cm is actually a good thing if, in the end, more space is needed for the spectroscopy part, or simply if something takes up more space than expected.

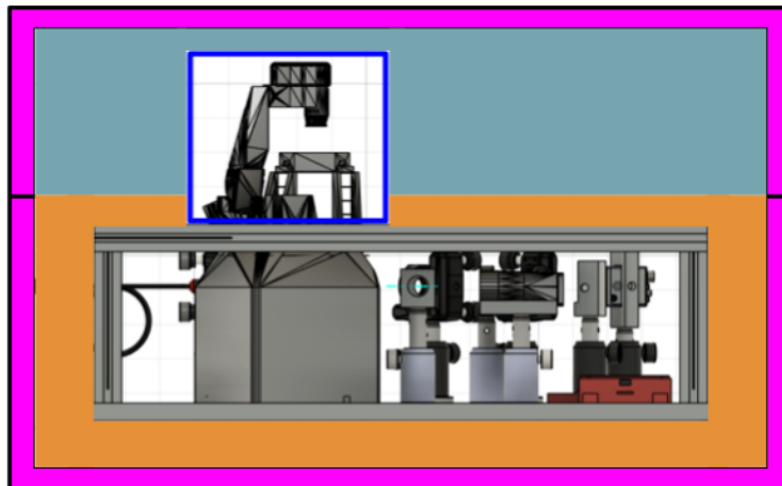


Figure 23: ThorLabs Breadboard 450 x 300 mm

Once the Raman is running on both the bench lab and the portable breadboard, it may be possible to optimize space even further and use a smaller breadboard.

8.2 Enclosure Design

The idea was to make a portable Raman microscope and put it in an expedition case (see Figure 25a). It's easy to transport, easy to store on a boat, quick access for the user and protects well against dust, moisture and shock.



- Closed suitcase for transport
- Top foam for shock absorption
- Bottom foam for protection
- Black box blocks light and impacts (removable for easy sample access)

Figure 24: Portable Raman Concept

Figure 24 illustrates the portable Raman concept. The breadboard will be placed on the case with foam around it. As shown in Figure 24, there will be a background of foam surrounding the microscope. This will not move at all, unless a repair needs to be made. On the other hand, the top foam will be fixed to the front of the case. So, when the case is opened, all you see is the black box covering the microscope and a little foam. The black box can be easily removed to guarantee easy access to the sample.



(a) Exemple of a suitable Suitcase



(b) Aluminium profile

For even better protection than foam alone, aluminum profiles (see Figure 25b) will be screwed directly to the breadboard around all optical elements, as shown in Figure 26.

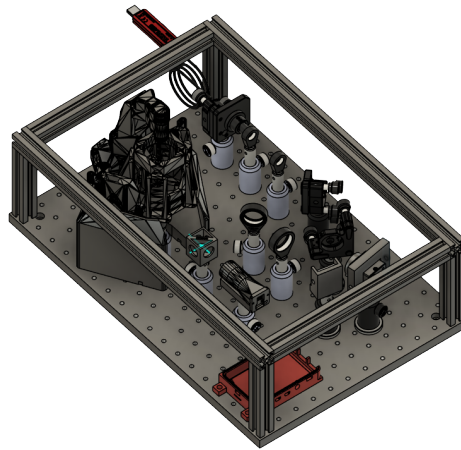


Figure 26: Fusion Assembly with the aluminum profile

Black polycarbonate panels will then be screwed directly onto the profile to protect the microscope from light. It also protects the optical element from dust and moisture. Polycarbonate is the most suitable material because it's a plastic and therefore better suited to the marine environment than wood, for example. What's more, it's highly UV-resistant, strong and durable over time, unlike other plastics such as PMMA.

To assemble aluminum profiles and polycarbonate panels, it's possible to buy commercial bolts and squares, but they can be quite expensive and heavy because there are so many of them. An alternative is to 3D print these parts, as shown in Figure 27 and Figure 28.



Figure 27: 3D printed Bolt for profile



Figure 28: 3D printed square for profile

9 Ongoing Improvements and Future Tasks

The project is still being pursued by two students for the summer, and here are the points they are already working on and will continue to work on:

- Finalizing the calibration of the photon camera: an accurate neon lamp spectrum needs to be collected for this, currently the use of a slit and/or telescope is being added to the path, this could help to get a smaller beam collimated before the grating and therefore an accurate neon lamp spectrum. The same photonic camera used in OpenRaman[9] is being used for our installation, and they are using a telescope, so this is the option that has been decided to explore for the time being.
- Rigorous calibration protocol for the photon camera: Once the calibration has been carried out, a rigorous protocol must be found to calibrate it correctly every time. This will be useful for future prototyping and use. Indeed, calibration may have to be carried out several times, as soon as an optical element moves, it has to be redone.
- Transfer to a portable setup and build the enclosure: Once the Raman is running on the benchtop lab, it needs to be transferred and run on the portable breadboard. With the final size of the Raman microscope, the enclosure can be built accordingly.
- Add the 3D-printed mount: Once everything is working, the 3D-printed mount can be reintroduced into the installation to see if it's a viable option. Posts and posts holders could also be replaced. If this works, it will considerably reduce the cost of the Raman microscope.

10 Conclusion

The aim of this semester's project was to design a portable, low-cost Raman microscope that could identify microplastics directly in the field — particularly on small boats, such as those used by the Sailowtech Association. Building on an earlier prototype, we identified its limitations and made significant progress towards creating a more functional, field-ready device.

Throughout the project, we gained a thorough understanding of the fundamental principles of Raman spectroscopy and successfully implemented them. We reworked the optical alignment, improved mechanical stability, and selected a more appropriate diffraction grating. These improvements increase the probability of detecting clear Raman signals. Although the photonic camera is still being calibrated, a solid foundation has been established for the future.

The next steps — final calibration, validation on real samples, and integration into a complete, portable unit — will bring us closer to achieving our goal of creating a viable, standalone Raman microscope for environmental research. Ultimately, this device could facilitate the more accessible and sustainable detection of microplastics in marine ecosystems.

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